

Лабораторная работа №16

Базовая защита от атак типа «brute force»

Комягин А.Н.

Российский университет дружбы народов, Москва, Россия

Цель работы

Получение навыков работы с программным средством **Fail2ban** для обеспечения базовой защиты сервера от атак типа «brute force» (полный перебор паролей).

Задачи работы

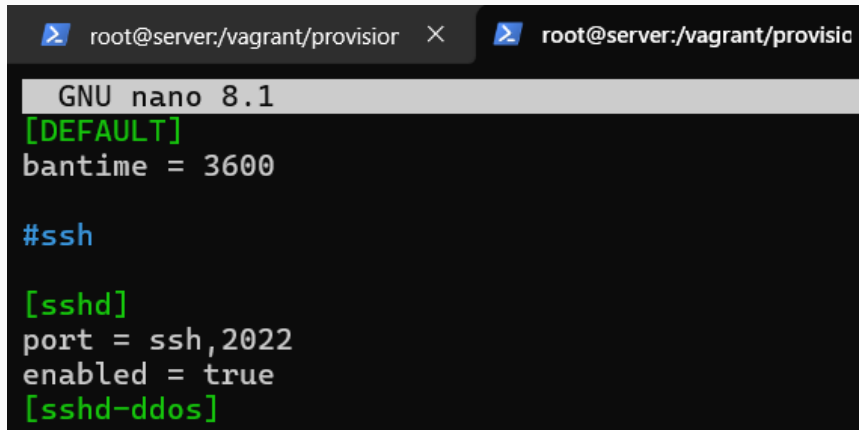
1. Установка и базовая настройка Fail2ban.
2. Настройка защиты служб SSH, Web (Apache) и Mail (Postfix/Dovecot).
3. Проверка работоспособности (имитация атаки).
4. Управление блокировками (бан/разбан, белые списки).
5. Автоматизация настройки.

Настройка Fail2ban

Конфигурация SSH

Создание локального файла конфигурации
`/etc/fail2ban/jail.d/customisation.local`.

Включение защиты для SSH и установка времени блокировки (`bantime`).



The screenshot shows a terminal window with two tabs. The active tab is titled `root@server:/vagrant/provisor`. The terminal content shows the `GNU nano 8.1` editor interface. The configuration file `/etc/fail2ban/jail.d/customisation.local` is being edited. The current content includes the `[DEFAULT]` section with `bantime = 3600`, the `#ssh` section header, and the `[sshd]` section with `port = ssh, 2022` and `enabled = true`. The `[sshd-ddos]` section header is also visible.

```
GNU nano 8.1
[DEFAULT]
bantime = 3600

#ssh

[sshd]
port = ssh, 2022
enabled = true
[sshd-ddos]
```

Запуск службы и проверка инициализации “тюрем” (jails) в логах.

```
root@server:/vagrant/provisic X root@server:/vagrant/provisior X + v
2025-11-15 17:07:47,705 fail2ban.database [30420]: INFO Connection to database closed.
2025-11-15 17:07:47,706 fail2ban.server [30420]: INFO Exiting Fail2ban
2025-11-15 17:07:48,559 fail2ban.server [31162]: INFO -----
2025-11-15 17:07:48,559 fail2ban.server [31162]: INFO Starting Fail2ban v1.1.0
2025-11-15 17:07:48,578 fail2ban.observer [31162]: INFO Observer start...
2025-11-15 17:07:48,605 fail2ban.database [31162]: INFO Connected to fail2ban persistent database '/var/lib/fail2ban/fail2ban.sqlite3'
2025-11-15 17:07:48,616 fail2ban.jail [31162]: INFO Creating new jail 'sshd'
2025-11-15 17:07:48,655 fail2ban.jail [31162]: INFO Jail 'sshd' uses systemd {}
2025-11-15 17:07:48,668 fail2ban.jail [31162]: INFO Initiated 'systemd' backend
2025-11-15 17:07:48,675 fail2ban.filter [31162]: INFO maxLines: 1
2025-11-15 17:07:48,749 fail2ban.filtersystemd [31162]: INFO [sshd] Added journal match for: '_SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-sessi
on'
2025-11-15 17:07:48,750 fail2ban.filter [31162]: INFO maxRetry: 5
2025-11-15 17:07:48,751 fail2ban.filter [31162]: INFO findtime: 600
2025-11-15 17:07:48,751 fail2ban.actions [31162]: INFO banTime: 3600
2025-11-15 17:07:48,752 fail2ban.filter [31162]: INFO encoding: UTF-8
2025-11-15 17:07:48,755 fail2ban.jail [31162]: INFO Creating new jail 'selinux-ssh'
2025-11-15 17:07:48,824 fail2ban.jail [31162]: INFO Jail 'selinux-ssh' uses pyinotify {}
2025-11-15 17:07:48,849 fail2ban.jail [31162]: INFO Initiated 'pyinotify' backend
2025-11-15 17:07:48,866 fail2ban.datedetector [31162]: INFO date pattern '': 'Epoch'
2025-11-15 17:07:48,870 fail2ban.filter [31162]: INFO maxRetry: 5
2025-11-15 17:07:48,871 fail2ban.filter [31162]: INFO findtime: 600
2025-11-15 17:07:48,872 fail2ban.actions [31162]: INFO banTime: 3600
2025-11-15 17:07:48,873 fail2ban.filter [31162]: INFO encoding: UTF-8
2025-11-15 17:07:48,897 fail2ban.filter [31162]: INFO Added logfile: '/var/log/audit/audit.log' (pos = 0, hash = 0908f3c298706a8f0cda6c27ee364104
4996201e)
2025-11-15 17:07:48,900 fail2ban.jail [31162]: INFO Creating new jail 'sshd-ddos'
2025-11-15 17:07:48,903 fail2ban.jail [31162]: INFO Jail 'sshd-ddos' uses pyinotify {}
2025-11-15 17:07:48,943 fail2ban.jail [31162]: INFO Initiated 'pyinotify' backend
2025-11-15 17:07:48,952 fail2ban.filter [31162]: INFO maxLines: 1
2025-11-15 17:07:48,958 fail2ban.filter [31162]: INFO maxRetry: 5
2025-11-15 17:07:48,959 fail2ban.filter [31162]: INFO findtime: 600
2025-11-15 17:07:48,960 fail2ban.actions [31162]: INFO banTime: 3600
2025-11-15 17:07:48,960 fail2ban.filter [31162]: INFO encoding: UTF-8
2025-11-15 17:07:48,979 fail2ban.jail [31162]: INFO Jail 'sshd' started
2025-11-15 17:07:49,008 fail2ban.jail [31162]: INFO Jail 'selinux-ssh' started
2025-11-15 17:07:49,017 fail2ban.jail [31162]: INFO Jail 'sshd-ddos' started
2025-11-15 17:07:49,054 fail2ban.filter [31162]: WARNING [selinux-ssh] Ignoring all log entries older than 600s; these are probably messages generat
ed while fail2ban was not running.
2025-11-15 17:07:49,059 fail2ban.filter [31162]: WARNING [selinux-ssh] Please check a jail for a timing issue. Line with odd timestamp: type=BPF msg
```


Добавление правил для Apache (auth, badbots, noscript) и почтовых сервисов.

```
# http

[apache-auth]
enabled = true
[apache-badbots]
enabled = true
[apache-noscript]
enabled = true
[apache-overflows]
enabled = true
[apache-nohome]
enabled = true
[apache-botsearch]
enabled = true
[apache-fakegooglebot]
enabled = true
[apache-modsecurity]
enabled = true
[apache-shellshock]
```

Проверка инициализации правил

В логах видно, что Fail2ban начал отслеживать логи веб-сервера и почты.

```
2025-11-15 17:11:19,474 fail2ban.server [32921]: INFO -----
2025-11-15 17:11:19,477 fail2ban.server [32921]: INFO Starting Fail2ban v1.1.0
2025-11-15 17:11:19,480 fail2ban.observer [32921]: INFO Observer start...
2025-11-15 17:11:19,543 fail2ban.database [32921]: INFO Connected to fail2ban persistent database '/var/lib/fail2ban/fail2ban.sqlite3'
2025-11-15 17:11:19,557 fail2ban.jail [32921]: INFO Creating new jail 'sshd'
2025-11-15 17:11:19,661 fail2ban.jail [32921]: INFO Jail 'sshd' uses systemd {}
2025-11-15 17:11:19,669 fail2ban.jail [32921]: INFO Initiated 'systemd' backend
2025-11-15 17:11:19,685 fail2ban.filter [32921]: INFO maxLines: 1
2025-11-15 17:11:19,749 fail2ban.filtersystemd [32921]: INFO [sshd] Added journal match for: '_SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-sessi
on'
2025-11-15 17:11:19,752 fail2ban.filter [32921]: INFO maxRetry: 5
2025-11-15 17:11:19,762 fail2ban.filter [32921]: INFO findtime: 600
2025-11-15 17:11:19,764 fail2ban.actions [32921]: INFO banTime: 3600
2025-11-15 17:11:19,764 fail2ban.filter [32921]: INFO encoding: UTF-8
2025-11-15 17:11:19,774 fail2ban.jail [32921]: INFO Creating new jail 'selinux-ssh'
2025-11-15 17:11:19,824 fail2ban.jail [32921]: INFO Jail 'selinux-ssh' uses pyinotify {}
2025-11-15 17:11:19,852 fail2ban.jail [32921]: INFO Initiated 'pyinotify' backend
2025-11-15 17:11:19,861 fail2ban.datedetector [32921]: INFO date pattern '': 'Epoch'
2025-11-15 17:11:19,862 fail2ban.filter [32921]: INFO maxRetry: 5
2025-11-15 17:11:19,862 fail2ban.filter [32921]: INFO findtime: 600
2025-11-15 17:11:19,862 fail2ban.actions [32921]: INFO banTime: 3600
2025-11-15 17:11:19,863 fail2ban.filter [32921]: INFO encoding: UTF-8
2025-11-15 17:11:19,863 fail2ban.filter [32921]: INFO Added logfile: '/var/log/audit/audit.log' (pos = 3497556, hash = 0908f3c298706a8f0cda6c27ee
3641044996201e)
2025-11-15 17:11:19,864 fail2ban.jail [32921]: INFO Creating new jail 'apache-auth'
2025-11-15 17:11:19,865 fail2ban.jail [32921]: INFO Jail 'apache-auth' uses pyinotify {}
2025-11-15 17:11:19,922 fail2ban.jail [32921]: INFO Initiated 'pyinotify' backend
2025-11-15 17:11:19,958 fail2ban.filter [32921]: INFO maxRetry: 5
2025-11-15 17:11:19,959 fail2ban.filter [32921]: INFO findtime: 600
2025-11-15 17:11:19,959 fail2ban.actions [32921]: INFO banTime: 3600
2025-11-15 17:11:19,960 fail2ban.filter [32921]: INFO encoding: UTF-8
2025-11-15 17:11:19,960 fail2ban.filter [32921]: INFO Added logfile: '/var/log/httpd/server.ankomyagin.net-error_log' (pos = 0, hash = )
2025-11-15 17:11:19,966 fail2ban.filter [32921]: INFO Added logfile: '/var/log/httpd/error_log' (pos = 577, hash = 889acb1cc01104e9b179d664eff7e3
1f222739ac)
2025-11-15 17:11:19,969 fail2ban.filter [32921]: INFO Added logfile: '/var/log/httpd/ssl_error_log' (pos = 171, hash = 787d78674eb01617b1aa446fe2
39606bd889a4a0)
2025-11-15 17:11:19,971 fail2ban.filter [32921]: INFO Added logfile: '/var/log/httpd/www.ankomyagin.net-error_log' (pos = 343, hash = c9c56aa456f
5213145eb1886c6762a076f527889)
2025-11-15 17:11:19,977 fail2ban.jail [32921]: INFO Creating new jail 'apache-badbots'
2025-11-15 17:11:19,979 fail2ban.jail [32921]: INFO Jail 'apache-badbots' uses pyinotify {}
2025-11-15 17:11:20,019 fail2ban.jail [32921]: INFO Initiated 'pyinotify' backend
```

Тестирование защиты

Изменение параметра `maxretry` (количество попыток) на 2 для быстрой проверки.

```
[root@server.ankomyagin.net client]# fail2ban-client status
Status
|- Number of jail:      16
|_ Jail list:  apache-auth, apache-badbots, apache-botsearch, apache-fakegooglebot, apache-modsecurity, apache-nohome, apache-noscript, apache-overflows,
apache-shellshock, dovecot, postfix, postfix-rbl, postfix-sasl, selinux-ssh, sshd, sshd-ddos
[root@server.ankomyagin.net client]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|_ - Currently failed: 0
|_ - Total failed:    0
|_ - Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
|- Actions
|_ - Currently banned: 0
|_ - Total banned:    0
|_ - Banned IP list:
[root@server.ankomyagin.net client]# fail2ban-client set sshd maxretry 2
2
[root@server.ankomyagin.net client]# |
```

Попытка подключения с клиента с неверным паролем. Результат: разрыв соединения (Connection refused) после превышения лимита.

```
[ankomyagin@client.ankomyagin.net ~]$ ssh vagrant@server.ankomyagin.net
vagrant@server.ankomyagin.net's password:
Permission denied, please try again.
vagrant@server.ankomyagin.net's password:
Permission denied, please try again.
vagrant@server.ankomyagin.net's password:

vagrant@server.ankomyagin.net: Permission denied (publickey,gssapi-keyex,gssapi-with-mic,password).
[ankomyagin@client.ankomyagin.net ~]$
[ankomyagin@client.ankomyagin.net ~]$ ssh vagrant@server.ankomyagin.net
ssh: connect to host server.ankomyagin.net port 22: Connection refused
[ankomyagin@client.ankomyagin.net ~]$ ssh vagrant@server.ankomyagin.net
ssh: connect to host server.ankomyagin.net port 22: Connection refused
[ankomyagin@client.ankomyagin.net ~]$ |
```

Управление блокировками

Статус и разблокировка

Просмотр заблокированных IP-адресов и ручная разблокировка клиента командой `unbanip`.

```
[root@server.ankomyagin.net client]# fail2ban-client status
Status
|- Number of jail:      16
|- Jail list:  apache-auth, apache-badbots, apache-botsearch, apache-fakegooglebot, apache-modsecurity, apache-nohome, apache-noscript, apache-overflows, a
apache-shellshock, dovecot, postfix, postfix-rbl, postfix-sasl, selinux-ssh, sshd, sshd-ddos
[root@server.ankomyagin.net client]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|  |- Currently failed: 0
|  |- Total failed:     0
|  '- Journal matches:  _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
'- Actions
|  |- Currently banned: 0
|  |- Total banned:     0
|  '- Banned IP list:
[root@server.ankomyagin.net client]# fail2ban-client set sshd maxretry 2
2
[root@server.ankomyagin.net client]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|  |- Currently failed: 0
|  |- Total failed:     4
|  '- Journal matches:  _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
'- Actions
|  |- Currently banned: 1
|  |- Total banned:     1
|  '- Banned IP list:  192.168.1.30
[root@server.ankomyagin.net client]# fail2ban-client set sshd unbanip 192.168.1.30
1
[root@server.ankomyagin.net client]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|  |- Currently failed: 0
|  |- Total failed:     4
|  '- Journal matches:  _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
'- Actions
|  |- Currently banned: 0
|  |- Total banned:     1
|  '- Banned IP list:
[root@server.ankomyagin.net client]# |
```

Добавление IP-адреса клиента в `ignoreip` для исключения из блокировок.

```
[root@server.ankomyagin.net client]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|  |- Currently failed: 0
|  |- Total failed:    0
|  '- Journal matches: _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
'- Actions
   |- Currently banned: 0
   |- Total banned:    0
   '- Banned IP list:
[root@server.ankomyagin.net client]# |
```


Контрольные вопросы

Принцип работы Fail2ban

Программа сканирует лог-файлы на наличие паттернов неудачных попыток входа и блокирует IP-адреса атакующих через Firewall (iptables/firewalld).

Приоритет конфигураций

Файлы с расширением `.local` имеют приоритет над файлами `.conf`.

Разблокировка IP

Используется команда клиента:

```
fail2ban-client set <jail_name> unbanip <IP-адрес>
```

Выводы

- Развернут сервис Fail2ban.
- Настроена защита ключевых служб (SSH, HTTP, SMTP/IMAP).
- Протестирован механизм автоматической блокировки злоумышленников.
- Изучены инструменты управления банами и белыми списками.